



Sri Chaitanya IIT Academy., India.

AP, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office , Madhapur – Hyderabad

Sec: Sr.Super60 & All

GTA-22

Date: 02-09-2021

Time: 02:30PM to 05:30

2018_P2

Max.Marks:180

02-09-2021_Sr.Super60 & All_Jee-Adv(2018-P2)_GTA-22_Syllabus

MATHS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

Name of the Student: _____

H.T. NO:

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JEE-ADVANCE-2018-P2-Model

Time: 3:00 Hour's

IMPORTANT INSTRUCTIONS

Max Marks: 180

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N : 7 – 14)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30)	+3	0	8	24
Sec – III(Q.N : 15-18)	Matrix Matching Type	+3	-1	4	12
Total				18	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N : 25 –32)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30)	+3	0	8	24
Sec – III(Q.N : 33-36)	Matrix Matching Type	+3	-1	4	12
Total				18	60

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N:37 – 42)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N :43–50)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30)	+3	0	8	24
Sec –III(Q.N : 51-54)	Matrix Matching Type	+3	-1	4	12
Total				18	60

PHYSICS

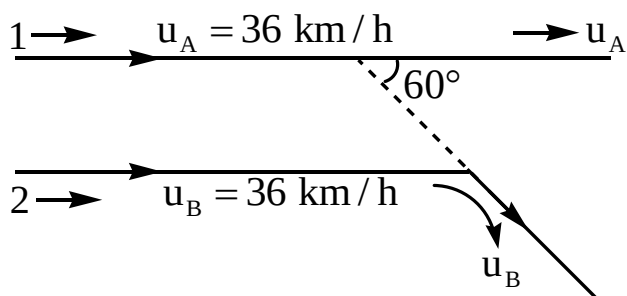
Max. Marks: 61

SECTION - I
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE than ONE** option can be correct.

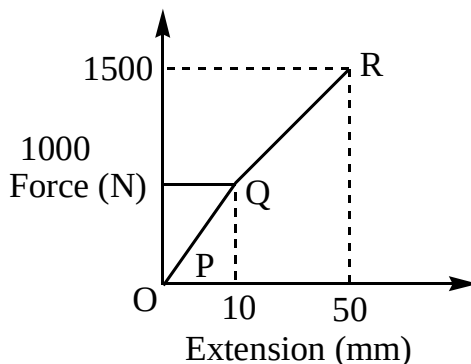
Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

1. Two trains move in the same direction on two close tracks. The train A sounds a horn of single frequency of 1 kHz. At a certain instant of time, when both trains move at the same speed of 36 km/hr, the two straight tracks deviate at an angle of 60° with each other. Velocity of sound in air is 300 m/s. Which of the following statement(s) is/ are true?

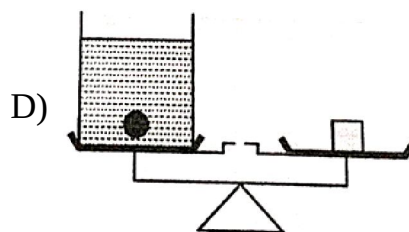
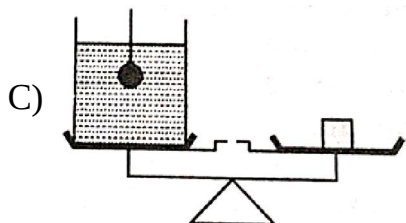
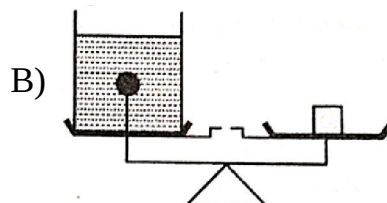
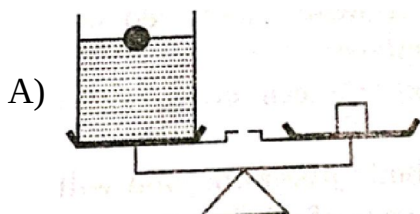


- A) The apparent frequency heard by a passenger on the second train B just after he passes the bend will be 1.05 kHz.
 B) The apparent frequency heard by a passenger on the first train A, just after the second train B passes the bend will be 1 kHz.
 C) The apparent frequency heard by a passenger on the second train B just after he passes the bend will be 983 Hz.
 D) If the first train A, that produces the sound were to pass the bend while the other train B goes on the straight track then the apparent frequency heard by a passenger on the latter train B will be less than 1 kHz.
2. Mark the **INCORRECT** options:
- A) If $\vec{d} - \vec{e} = \vec{f}$ and $f = d + e$ then $\vec{d}$ and $\vec{e}$ are opposite in direction
 B) $\vec{d} + \vec{e} = \vec{f}$ and $f = \sqrt{2}d; d = e$ then $\vec{d}$ and $\vec{e}$ are perpendicular
 C) If $\vec{d} - \vec{e} = \vec{f}$ and $f = d + e$ then $\vec{d}$ and $\vec{e}$ are in the same direction
 D) If $\vec{d} + \vec{e} = \vec{f}$ and $f = \sqrt{2}d; d = e$ then $\vec{d}$ and $\vec{e}$ are in opposite direction

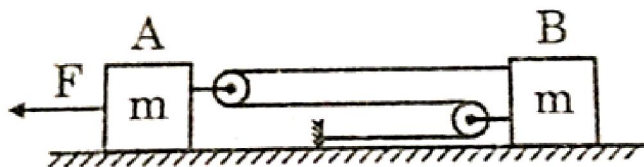
3. The graph shows the force-extension curve for a steel wire that is stretched by a tensile force. In the graph, PQ and QR are two idealized straight line sections. The unextended length of the wire is 2.00 m. Assume that Q is proportionality limit and R is yield point. Then mark the **CORRECT** statement(s).



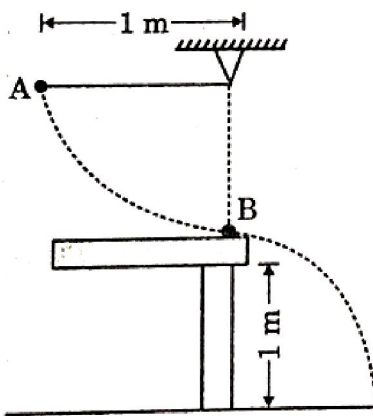
- A) The strain in the wire at R is 0.025
 B) The work done in slowly stretching the wire from Q to R is equal to 50 J
 C) If we release the wire at point Q, it won't have any permanent set
 D) The work done in slowly stretching the wire from Q to R is equal to 55 J
4. With a wooden ball and a high water tank we hold four experiments. In the first experiment we have a floating ball weighed in a vessel. In the second experiment, the ball is weighed while tied to the bottom of the vessel, the third time the ball held under water by using the thin rod, finally, during the fourth experiment the ball is taken deep inside and released. In all the four cases weight is recorded. In which case the weight recorded is more than the weight of ball + water + tank?



5. Two blocks A and B of equal mass are connected by a light inextensible taut string passing over two light smooth pulleys fixed to the blocks. The parts of the string not in contact with the pulleys are horizontal. A horizontal force F is applied to the block A as shown. There is no friction, then:



- A) The acceleration of A will be more than that of B
 B) The acceleration of A will be less than that of B
 C) The sum of rate of changes of momentum of A and B is greater than the magnitude of F
 D) The sum of rate of changes of momentum of A and B is equal to the magnitude of F
6. A small steel ball B is at rest on the edge of a table of height 1 m. Another steel ball A, used as the bob of a meter long simple pendulum, is released from rest with the pendulum suspension horizontal, and end swings against B as shown in the figure. The masses of the balls are identical and the collision is elastic. Consider the motion of B only up until it first hits the ground.



- A) Ball A is in motion for longer time
 B) Ball B is in motion for longer time
 C) Ball A has greater path length
 D) Ball B has greater path length

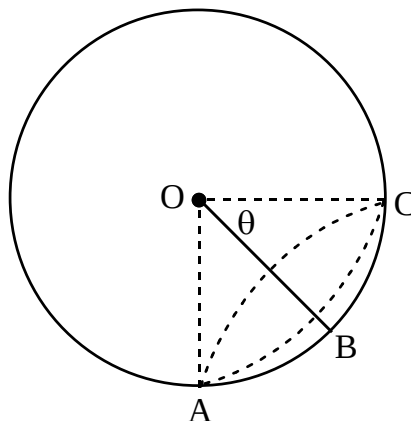
SECTION-II
(NUMERICAL VALUE TYPE)

This section contains 8 questions. Each question is numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place.

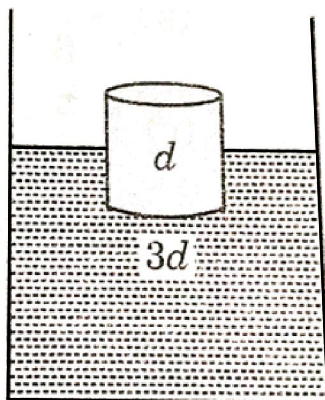
(e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.

7. End A of a rod AB of length $L = 0.5 \text{ m}$ and of uniform cross-sectional area is maintained at some constant temperature. The heat conductivity of the rod is $K = 11.34 \text{ W m}^{-1} \text{ K}^{-1}$. The other end B of this rod is radiating energy into vacuum and the wavelength with maximum energy density emitted from this end is $\lambda_0 = 75000 \text{ \AA}$. If the emissivity of the end B is $e = 1$, determine the temperature of the end A (in Kelvin). Assume that lateral surface is well lagged. (Take Wien's constant $= 3 \times 10^{-3} \text{ mK}$, Stefan's constant $\sigma = 5.67 \times 10^{-8} \text{ W / m}^2 \text{ K}^4$)
8. There are two layers of water in a calorimeter, the lower one colder at temperature 30°C and the upper one hotter at temperature 60°C . The mass of lower layer water is m and volume is V . The mass of upper layer water is $2m$ and volume is $2V$. The coefficient of volume expansion for water is γ . When the temperature of the whole system becomes uniform, the ratio of new volume to the old volume of the entire system is: _____ ?
9. Consider a small water drop of radius R in air. If T is the surface tension, then the force due to surface tension acting on the smaller section ABC is F . Line OB is line of symmetry to section ABC with angle $\angle BOC = \angle BOA = 30^\circ$. Find the value of $\frac{\pi TR}{F}$?

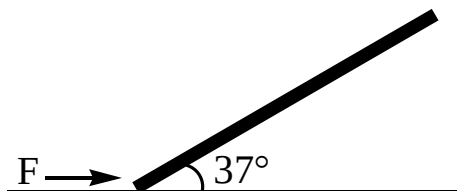


10. A cylindrical block of height 1 m is in equilibrium in a beaker as shown. Cross-sectional area of cylindrical block is one fourth of cross-sectional area of beaker. Density of cylindrical block is one third of liquid. Determine the time period of small oscillation (in seconds). (Given : $g = \pi^2 \text{ m/s}^2$) (Ignore surface tension and viscosity effects)



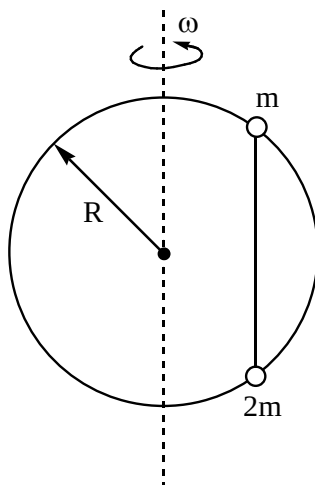
11. A rod is pushed on a smooth horizontal floor and moves in a vertical plane as shown in figure. During motion, rod maintains a constant angle 37° with horizontal. If mass of rod is 300 gms, find the value of F (in N) required to do so.

(Take $g = 10 \text{ m/s}^2$, $\sin 37^\circ = \frac{3}{5}$)

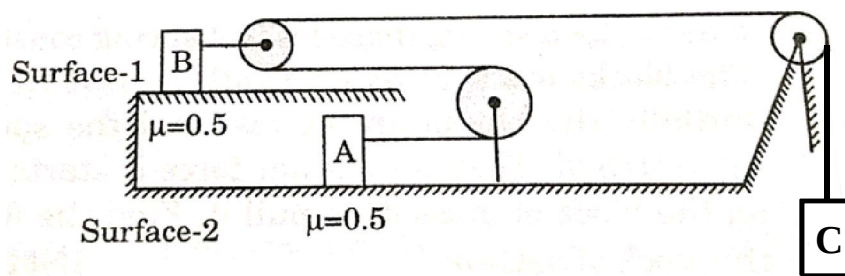


12. Two beads of mass m and $2m$ are connected by a massless rod of length R and threaded onto a smooth circular wire of radius R . When the wire rotates uniformly with angular velocity ω about a vertical diameter, the beads do not slide on it and

rod remains vertical as shown. What must be the angular velocity ω (in rad/s) of the wire ? (Take $R = 15$ m)



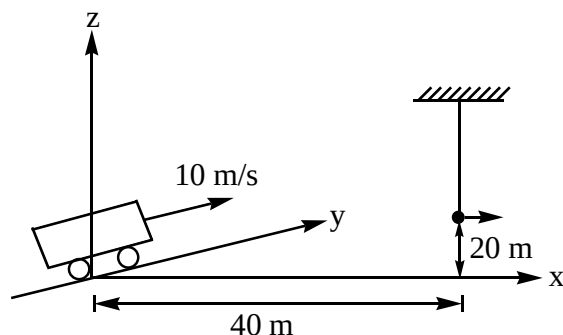
13. Three blocks A, B, C, whose masses are 9 kg, 9 kg and 18 kg respectively as shown in the figure, are released from rest. The coefficient of friction between block A and surface-2 is 0.5 and the coefficient of friction between block B and surface-1 is also 0.5. The acceleration of block C just after release is $\frac{n \times 10}{9} (\text{m/s}^2)$. All strings and pulleys are ideal. Find the value of n ?



14. A pendulum bob is in x-z vertical plane. A cart is moving along y-axis with velocity 10 m/s from which a ball is to be projected such that it strikes pendulum bob with minimum speed while travelling horizontally in x-z plane. Ball is projected from origin of coordinates. Pendulum bob is located at horizontal distance 40 m from point of projection above x-axis. If ball is projected with

velocity $v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$ relative to cart the find the value (in m/s) of sum

$$\frac{|v_x| + |v_y| + |v_z|}{10}.$$

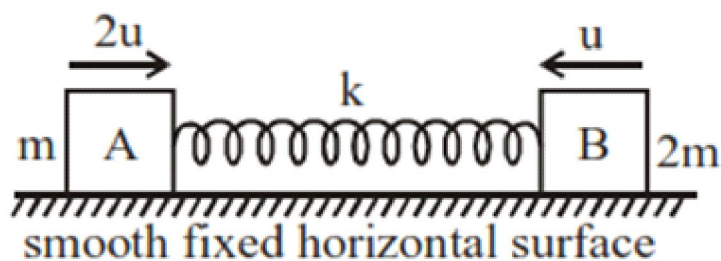


SECTION-III (MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

15. Two blocks A and B of mass m and $2m$ connected by a light ideal spring of spring constant k lie at rest on a fixed smooth horizontal surface. Initially the spring is unstretched. Now at time $t = 0$ both the blocks are imparted horizontal velocities towards each other of magnitudes $2u$ and u as shown in figure. In the subsequent motion, the only horizontal force acting on blocks is due to spring. Match the conditions in list-I with the instants of time they occur as given in list-II.



	List – I		List – II
(P)	The length of spring is least at time	(1)	$t = \frac{\pi}{2} \sqrt{\frac{2m}{3k}}$
(Q)	The lengths of spring is maximum at time	(2)	$t = \pi \sqrt{\frac{2m}{3k}}$
(R)	The acceleration of block A is zero at time	(3)	$t = \pi \sqrt{\frac{3m}{2k}}$
(S)	Velocity of block B is maximum at time	(4)	$t = \frac{\pi}{2} \sqrt{\frac{3m}{2k}}$

Codes :

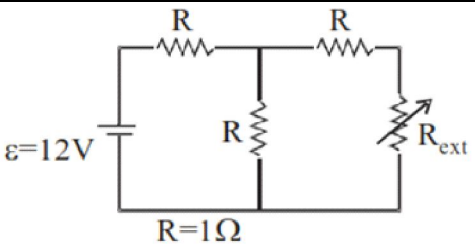
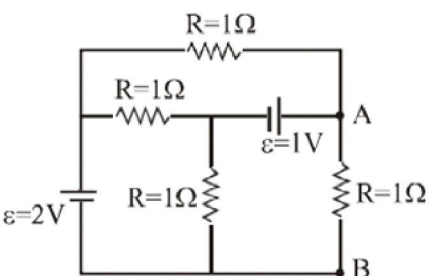
A) P → 1; Q → 3; R → 2; S → 4

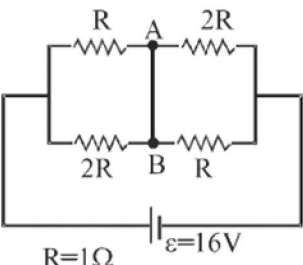
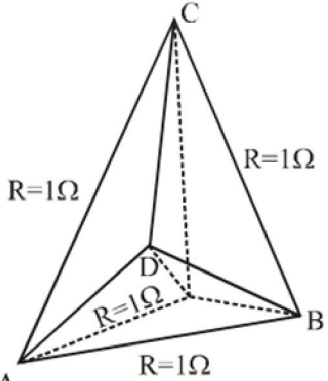
B) P → 1; Q → 3; R → 2; S → 2

C) P → 3; Q → 1; R → 4; S → 2

D) P → 1; Q → 1; R → 2; S → 2

16. Match the list-I with list-II.

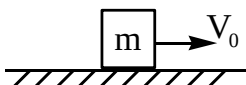
	List – I		List – II
(P)	 Numerical value of maximum power (in watt) delivered to R_{ext} is	(1)	$\frac{1}{2}$
(Q)	 Numerical value of potential difference (in volt) between points A & B is	(2)	$\frac{3}{2}$

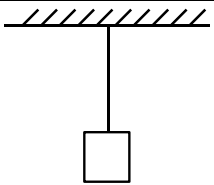
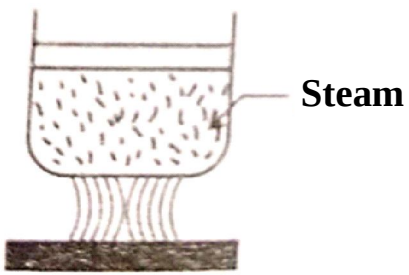
(R)	 Wire AB has negligible resistance. Numerical value of current (in ampere) through AB is	(3)	6
(S)	 Six resistances are arranged in tetrahedron (shown in solid lines) configuration. All resistance are equal. Numerical value of equivalent resistance (in Ω) between A & B is	(4)	4

Codes :

- A) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$ B) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$
 C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$ D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

17. In column-I, statement about heat given to the system, work and internal energy are given. Symbols have their usual meaning use first law of the thermodynamics in form $\Delta Q = \Delta U + \Delta W$. Match the situations in column-II with statements in column-I.

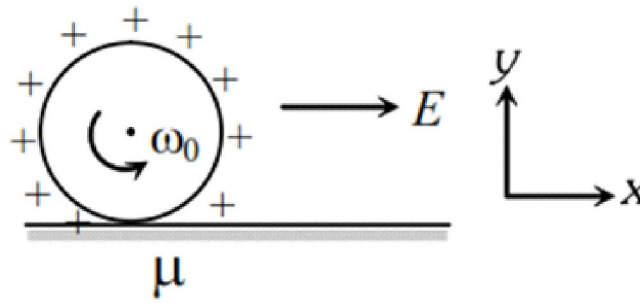
	List – I		List – II
(P)	 Rough ground Block is the system. Assume no heat is lost to surrounding till the block comes to rest.	(1)	$\Delta Q > 0$

(Q)	 A block is released from rest from mass less ideal non-conducting elastic string at natural length. After a few oscillations, block comes to rest. The string is the system. Assume the whole set-up is in vacuum. All joints insulated.	(2)	$\Delta Q < 0$
(R)	 Water is slowly boiled in a closed container with a movable light piston. The container with its contents is the system. (Excluding piston) Don't ignore atmosphere pressure	(3)	$\Delta W > 0$
(S)	A monatomic ideal gas undergoes a process $PT^3 = \text{constant}$. Gas is the system. Pressure increase during the process.	(4)	$\Delta U > 0$
		(5)	$\Delta U = 0$

Codes :

A) $P \rightarrow 4$; $Q \rightarrow 3$; $R \rightarrow 1, 3$; $S \rightarrow 2$ B) $P \rightarrow 3$; $Q \rightarrow 4$; $R \rightarrow 1, 4$; $S \rightarrow 2$ C) $P \rightarrow 4$; $Q \rightarrow 4$; $R \rightarrow 1, 3, 4$; $S \rightarrow 2$ D) $P \rightarrow 4$; $Q \rightarrow 4$; $R \rightarrow 1, 4$; $S \rightarrow 3$

18. A uniformly charged non-conducting hollow sphere of radius 1m and mass 2kg having angular speed $\omega_0 = 9 \text{ rad/s}$ is gently placed on non-conducting rough horizontal surface with coefficient of friction μ at $t = 0$. Charge on the sphere is Q_0 and uniform electric field E is present in the region parallel to horizontal surface. (Given $Q_0 E = 10 \text{ N}$, $g = 10 \text{ m/s}^2$).



	List – I		List – II
(P)	If $\mu = 0.6$, then a_{cm} till $t = 1$ sec	(1)	Along +ve x-axis
(Q)	If $\mu = 0.4$, then a_{cm} till $t = 1$ sec	(2)	Along -ve x-axis
(R)	If $\mu = 0.5$, then a_{cm} till $t = 1$ sec	(3)	Zero
(S)	If $\mu = 0.6$, then friction force till $t = 1$ sec	(4)	For some time along +ve x-axis and for some time along (-ve) x-axis

Codes :

A) $P \rightarrow 2$; $Q \rightarrow 1$; $R \rightarrow 3$; $S \rightarrow 2$

B) $P \rightarrow 1$; $Q \rightarrow 2$; $R \rightarrow 3$; $S \rightarrow 2$

C) $P \rightarrow 2$; $Q \rightarrow 1$; $R \rightarrow 2$; $S \rightarrow 3$

D) $P \rightarrow 3$; $Q \rightarrow 2$; $R \rightarrow 1$; $S \rightarrow 4$

CHEMISTRY

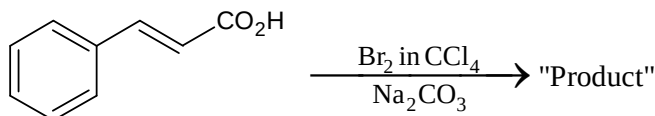
Max.Marks:61

SECTION - I
(MULTIPLE CORRECT ANSWER TYPE)

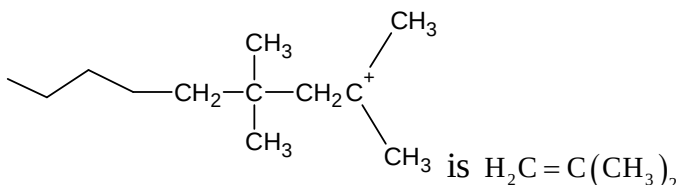
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE than ONE** option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

19. Which of the following statement is/are incorrect for the given reaction?



- A) Product does not show geometrical isomerism.
 B) Molecular formula of the product is $C_9H_8O_2Br_2$.
 C) Product shows optical isomerism.
 D) In total four stereoisomeric structures are possible for the product.
20. Which is/are correct statement?



- A) One of the monomer of
 B) Terylene is a thermosetting polymer
 C) Nylon-2-nylon-6 is a biodegradable polymer
 D) Natural rubber is all cis-polyisoprene
21. The correct statement(s) among the following is/are
- A) Cu^{2+} (aq) ions react with $K_4[Fe(CN)_6]$ to form chocolate brown precipitate.
 B) Cu^{2+} (aq) ions react with $K_3[Fe(CN)_6]$ to form green precipitate.
 C) Hg_2Cl_2 when treated with NH_4OH gives a black residue.
 D) NH_4Cl solution when treated with Nessler's reagent gives green precipitate.
22. Which of the following ore(s) is/are concentrated by froth floatation process
- A) Galena
 B) Copper pyrite
 C) Zinc blende
 D) Bauxite

23. In wave mechanical model, wave function of orbitals is represented by ψ_{n,ℓ,m_ℓ} .

Based on the following given wave function of orbitals

$$\psi_{1,0,0} = \frac{2}{a_0^{3/2}} e^{-r/na_0}$$

$$\psi_{3,0,0} = \frac{1}{9\sqrt{3}} \left(\frac{1}{a_0} \right)^{3/2} (6 - 6\sigma + \sigma^2) e^{-\sigma/2}, \text{ where } \sigma = \frac{2r}{na_0}$$

$$\psi_{2,0,0} = \frac{1}{2\sqrt{2}} \left(\frac{1}{a_0} \right)^{3/2} \left(2 - \frac{2r}{na_0} \right) e^{-r/na_0}$$

$$\psi_{3,1,0} = \frac{1}{9\sqrt{6}} \left(\frac{1}{a_0} \right)^{3/2} (4 - \sigma) \sigma e^{-\sigma/2}, \text{ where } \sigma = \frac{2r}{na_0}$$

Select the correct statement(s)

- A) For 3p orbital, the graph will intersect r-axis at two distinct point.
 B) For 1s orbital, of H-atom, the radial probability of finding electron will be maximum at $r = a_0$.
 C) For 3s orbital, number of angular node is zero.
 D) For 2s orbital, radial distance for radial node is $2a_0$
24. An aqueous solution of Sodium Sulphate was electrolysed for 19.3 minutes. 2.24 lt of oxygen gas is produced at the anode over water at 273.15 K and at a total pressure of 400 torr. If the vapour pressure of water at 273.15 K is 20 torr, then which of the following is/are correct: (1 F = 96500 C)
- A) the constant current used in electrolysis is 16.67 ampere.
 B) the pressure of oxygen gas over water at anode is 0.5 atm.
 C) the moles of hydrogen gas obtained at cathode is 0.1.
 D) the pH of solution after electrolysis is 7.

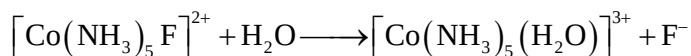
SECTION-II (NUMERICAL VALUE TYPE)

This section contains 8 questions. Each question is numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place.
 (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.

25. The compression factor (compressibility factor) for one mole of a vander Waals gas at 0°C and 100 atmosphere pressure is found to be 0.5. Assuming that the volume of a gas molecule is negligible, calculate the vander waals constant 'a' ($\text{atm l}^2 \text{mol}^{-2}$).

26. The complex $[\text{Co}(\text{NH}_3)_5\text{F}]^{2+}$ reacts with water according to the equation.

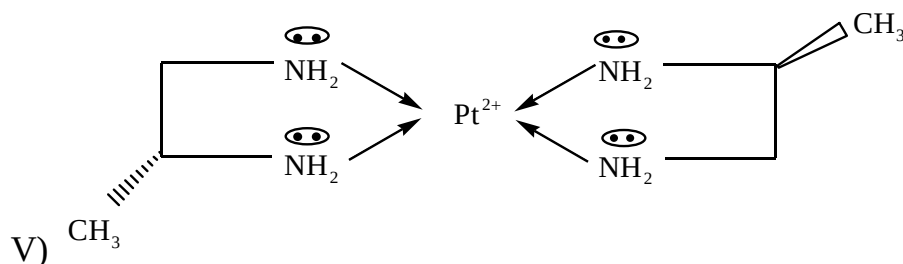
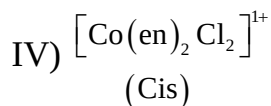
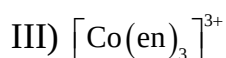


The rate of the reaction = $\text{rate const.} \times [\text{complex}]^a \times [\text{H}^+]^b$. The reaction is acid catalysed i.e. $[\text{H}^+]$ does not change during the reaction. Thus rate = $k'[\text{Complex}]^a$ where $k' = k[\text{H}^+]^b$, calculate 'a' and 'b' given the following data at 25°C. Represent your answer as (a+b)

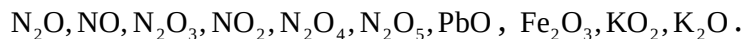
$[\text{Complex}]\text{M}$	$[\text{H}^+]\text{M}$	$T_{1/2}(\text{hr})$	$T_{3/4}(\text{hr})$
0.1	0.01	1	2
0.2	0.02	0.5	1

27. 24 gm of carbon and 48 gm of oxygen reacts completely to form $\text{CO}(\text{g})$ and $\text{CO}_2(\text{g})$. Calculate density of gaseous mixture (in gm per litre) at 1.642 atm pressure and 360 K temperature.

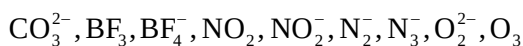
28. How many of the given species are chiral (Assuming these compound exist in nature)



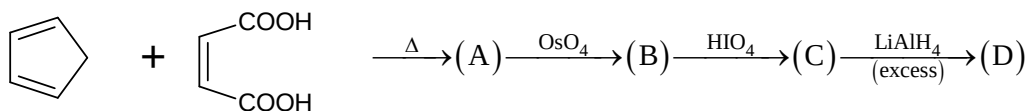
29. How many of the following oxides are colored?



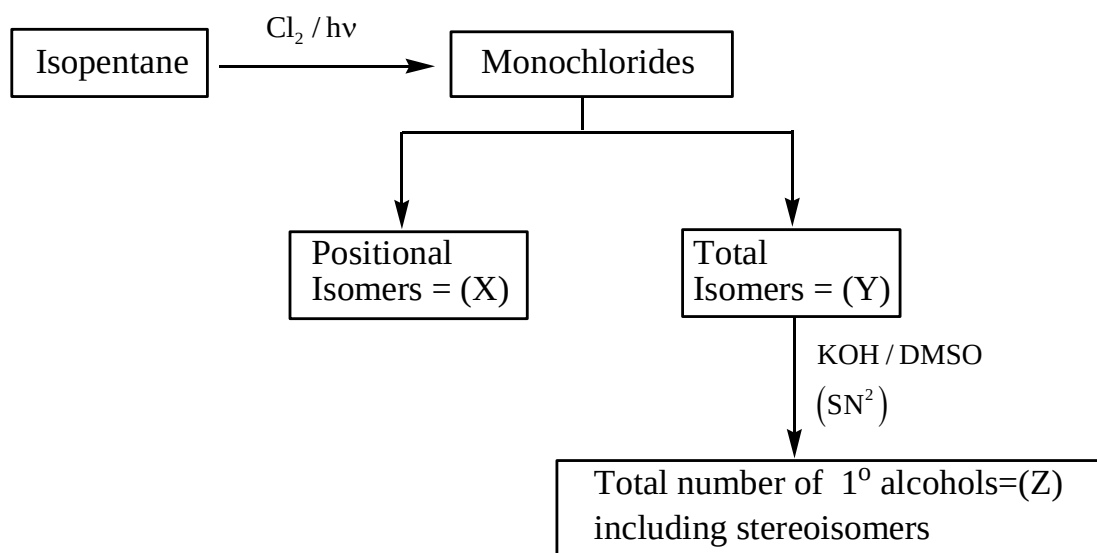
30. How many of the following species have Bond order greater than 1.33?



31. In the following sequence of reactions, how many hydroxyl groups ($-\text{OH}$) are present in the final product (D)?



32. Observe the following flow chart and answer the number of x, y, z.



Note: If value of $x = 1$; $y = 2$; $z = 3$. Then your answer is 123.

SECTION-III (MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

33. Given List I different ionic compounds and List II have their properties.

	List – I		List – II
(I)	KO ₂	(P)	Paramagnetic
(II)	K ₂ O ₂	(Q)	Diamagnetic
(III)	K ₂ O	(R)	Aqueous alkaline solution
(IV)	KO ₃	(S)	CO ₂ absorber and O ₂ releaser
		(T)	Coloured

Which of the following options has the correct combination considering List-I and List – II?

- A) (IV), (Q) B) (I), (T) C) (III), (S) D) (III), (P)

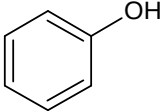
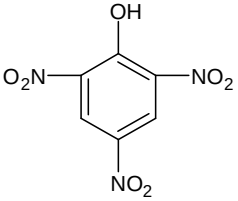
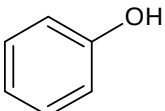
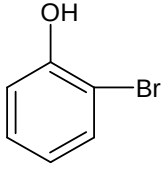
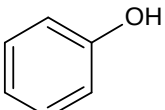
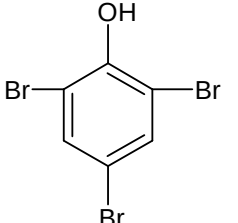
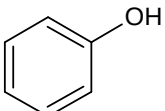
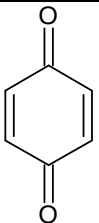
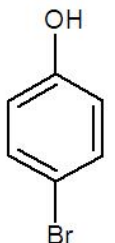
34.

	List – I		List – II
(P)	A compound that liberates ammonia on treatment with hot aqueous KOH	(1)	CH ₃ – CH ₂ – CN
(Q)	A compound that gives CO ₂ , NH ₄ ⁺ and acetic acid after boiling in conc. aqueous HCl	(2)	CH ₃ – CH ₂ – CONH ₂
(R)	A compound that gives NH ₄ ⁺ and mono carboxylic acid after treatment with concentrated aqueous HCl.	(3)	NC – CH ₂ – CN
(S)	A compound that gives product A with MeMgI followed by hydrolysis, which on further treatment with MeMgI excess followed by acidification. Gives 2-methyl-2-butanol	(4)	CH ₃ COONH ₄
		(5)	PhCOONH ₄

The correct option is _____

- | | | | | |
|----|-----------|-----|---------|---|
| | P | Q | R | S |
| A) | 1,3,5 | 1,3 | 1,4,5 | 3 |
| B) | 1,2,3,4,5 | 3 | 1,2,4,5 | 1 |
| C) | 3,4,5 | 4 | 1,5 | 2 |
| D) | 2,4 | 2 | 1,3,5 | 1 |

35. List type Question

	Column-I		Column – II
a)	 $\xrightarrow[\text{[H}^+]{\text{K}_2\text{Cr}_2\text{O}_7}$	(P)	 Picric acid
b)	 $\xrightarrow{\text{Br}_2 \text{ water}}$	(Q)	 Ortho – bromophenol
c)	 $\xrightarrow{\text{Conc. HNO}_3}$	(R)	 Tri – bromophenol
d)	 $\xrightarrow[\text{CS}_2(\ell)]{\text{Br}_2}$	(S)	 Para – benzoquinone
		(T)	 Para - bromophenol

Code:

A) a-qa; b-sr; c-ps; d-rt

B) a-s; b-pr; c-pq; d-st

C) a-s; b-r; c-ps; d-qs

D) a-s; b-r; c-p; d-qt;

36.

	List – I		List-II
(P)	$a = b \neq c, \alpha = \beta = 90^\circ, \gamma = 120^\circ$	(1)	Tetragonal
(Q)	$a \neq b \neq c, \alpha = \beta = \gamma = 90^\circ$	(2)	Monoclinic
(R)	$a \neq b \neq c, \alpha = \gamma = 90^\circ, \beta \neq 90^\circ$	(3)	Orthorhombic
(S)	$a = b \neq c, \alpha = \beta = \gamma = 90^\circ$	(4)	Hexagonal

A) P-4; Q-3; R-2; S-1

B) P-4; Q-2; R-3; S-1

C) P-1; Q-2; R-3; S-4

D) P-2; Q-3; R-1; S-4

MATHS**Max.Marks:61****SECTION - I****(MULTIPLE CORRECT ANSWER TYPE)**

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE than ONE** option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

37. Let $B = [b_{ij}]_{n \times n}$ be a square matrix of order n such that $b_{ij} = \begin{cases} 0 & i \neq j \\ \frac{4+i+j}{2}, & i = j \end{cases}$

A) $\lim_{m \rightarrow \infty} \left(\frac{\sum_{n=1}^m \text{trace}(B)}{m^3} \right) = \frac{1}{6}$

B) $\sum_{n=1}^{10} \text{trace}(B) = 320$

C) $\lim_{m \rightarrow \infty} \left(\frac{\sum_{n=1}^m \text{trace}(B)}{m^3} \right) = \frac{1}{3}$

D) $\sum_{n=1}^{10} \text{trace}(B) = 330$

38. If a pair of variable straight lines $x^2 + 4y^2 + \alpha xy = 0$ (where α is a real parameter) cut the ellipse $x^2 + 4y^2 = 4$ at two points A and B, then the locus of the point of intersection of tangents at A and B is

A) $x^2 - 4y^2 + 8xy = 0$

B) $(2x - y)(2x + y) = 0$

C) $x^2 - 4y^2 + 4xy = 0$

D) $(x - 2y)(x + 2y) = 0$.

39. A(1, 1, 1), B(1, 0, 0), C(3, 0, 0) are 3 vertices of a tetrahedron whose 4th vertex is D. The perpendicular drawn from D on to opposite face ABC meets the median line through A of triangle ABC at E. if AD = 4 and volume of tetrahedron is $\frac{2\sqrt{2}}{3}$ then

A) The length of altitude from D to opposite face ABC is 2

B) Only one E exist i.e. E (-1, 3, 3).

C) There exist two positions for E : (-1, 3, 3) and (3, -1, -1).

D) $\vec{j} - \vec{k}$ is normal to the plane of A, B, C

40. $f(x)$ and $g(x)$ are two functions which satisfy differential equation

$$\frac{d^2x}{dy^2} \frac{d^2y}{dx^2} + \frac{dx}{dy} = 0 \text{ where } f(0) = g(0) = 2 \text{ and } f'(0) = g'(0) = 1, \text{ then which of the}$$

following is/are **CORRECT**?

A) Both $y = f(x)$ and $y = g(x)$ are decreasing functions

B) both $y = f(x)$ and $y = g(x)$ are monotonic functions

C) Area bounded by $y = f(x)$, $y = g(x)$ and $x = \ln 2$ is $\left(\frac{3}{2} + \ln 4\right)$ sq. units

D) Area bounded by $y = f(x)$, $y = g(x)$ and $x = \ln 2$ is $\left(\frac{3}{2} - \ln 4\right)$ sq. units

41. Consider $f : [1, \infty) \rightarrow \mathbb{R}$ is given by $f(x) = (\log_e x)^2 - \int_1^x \frac{f(t)}{t} dt$, then which of the

following is/are **CORRECT**?

A) $f(e) = \frac{5}{6}$

B) $f(x) \geq -1 \forall x \in [1, \infty)$

C) area of the figure bounded by the tangent line of $y = f(x)$ at point $(e, f(e))$, the curve $y = f(x)$ and line $x = 1$ is $e - \frac{1}{e} + 3$

D) area of the figure bounded by the tangent line of $y = f(x)$ at point $(e, f(e))$, the curve $y = f(x)$ and line $x = 1$ is $e + \frac{1}{e} - 3$.

42. Let f be twice differentiable real valued function satisfying

$$f'(x) \neq 0, f(x) + f''(x) = -xg(x)f'(x)$$

Where $g(x) > 0 \forall x > 0$. If $f(0) = -3$ and $f'(0) = 4$ then

A) $|f(x)| \leq 5 \forall x > 0$

B) $(f(x))^2 + (f'(x))^2$ is decreasing $\forall x > 0$

C) $|f(x)| > 5 \forall x > 0$

D) $(f(x))^2 + (f'(x))^2$ is increasing $\forall x > 0$

SECTION-II

(NUMERICAL VALUE TYPE)

This section contains 8 questions. Each question is numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place.

(e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.

43. If sum of maximum and minimum value of

$$y = \log_2(x^4 + x^2 + 1) - \log_2(x^4 + x^3 + 2x^2 + x + 1)$$

can be expressed in form $((\log_2 m) - n)$, where m and 2 are coprime then value of $(m + n)$

44. When rolling a certain unfair six sided die with faces numbered 1, 2, 3, 4, 5 and 6, the probability of obtaining a specified face F is greater than $\frac{1}{6}$ and the probability of the face opposite to it is less than $\frac{1}{6}$. The probability of obtaining other four faces are each equals to $\frac{1}{6}$, and the sum of the numbers on opposite faces is 7.

When two such dice are rolled, the probability of obtaining a sum of 7 is $\frac{47}{288}$. If

the probability of obtaining the face F is $\frac{m}{n}$, where m and n are relatively prime positive integers, then value of $(m + n)$ is _____

45. Number of solution(s) of equation

$$(\cos^5 x) + (\sin x)(\cos^4 x) - (\sin^4 x) + (\sin^3 x) + (\cos x)(\sin^2 x) - (\cos^2 x) = 0$$

in $x \in [0, \pi]$ is _____

46. Two circles of radii $\sqrt{3}$ and 1 cut orthogonally, then area of common region is A. Value of A upto two decimal is _____ [take: $\pi \approx 3.14$, $\sqrt{3} \approx 1.73$]

47. Bunny the triangle is playing in the (x, y) plane. Let his maximum y coordinate be k. Given that he has side lengths 6, 8 and 10 and that no part of him is below the x-axis, minimum possible value of k is _____

48. If the distance of two points P and Q on parabola $y^2 = 4ax$ ($a > 0$) from the focus of a parabola are 4 and 9, respectively then the distance of the point of intersection of tangents at P and Q from the focus is
49. In a triangle PQR, let $\vec{a} = \overrightarrow{QR}$, $\vec{b} = \overrightarrow{RP}$ and $\vec{c} = \overrightarrow{PQ}$. If $|\vec{a}| = 3, |\vec{b}| = 4$ and $\frac{\vec{a} \cdot (\vec{c} - \vec{b})}{\vec{c} \cdot (\vec{a} - \vec{b})} = \frac{|\vec{a}|}{|\vec{a}| + |\vec{b}|}$, then the value of $|\vec{a} \times \vec{b}|^2$ is _____
50. $\prod_{k=1}^{2020} \left\{ 1 + 2 \cos \left(\frac{2\pi \cdot 3^k}{3^{2020} + 1} \right) \right\} =$

SECTION-III

(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

51.

	Column I		Column II
(P)	If $\alpha, \beta \in (0, \pi)$ and $\alpha \neq \beta$ satisfy the equation $\frac{1 - \cos 2\theta}{\sin \theta} = \frac{1}{2}$, then the value of $\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right)$ is equal to	(1)	0
(Q)	If the expression $\frac{x^2 + (2m+3)x + (m^2+3)}{\sqrt{x^2 + (2m+1)x + m^2 + 2}}$ is non-negative $\forall x \in \mathbb{R}$, then m can be equal to	(2)	8
(R)	Possible integral value(s) of x for which the parabola $y = 5x^2 + x - 3$ lies above the parabola $y = 2x^2 + 6x - 1$	(3)	1
(S)	The number of real solutions of the equation $x^{2 \log_x(x+3)} = 16$ is equal to	(4)	-1
		(5)	2

A) P $\rightarrow$ 4; Q $\rightarrow$ 2; R $\rightarrow$ 1; S $\rightarrow$ 3B) P $\rightarrow$ 2; Q $\rightarrow$ 3; R $\rightarrow$ 1; S $\rightarrow$ 2C) P $\rightarrow$ 4; Q $\rightarrow$ 1; R $\rightarrow$ 3; S $\rightarrow$ 2D) P $\rightarrow$ 2; Q $\rightarrow$ 4; R $\rightarrow$ 2; S $\rightarrow$ 1

52. Consider a sequence $\{b_n\}$ of integers such that b_1, b_2, b_3 are in G.P. b_2, b_3, b_4 are in A.P., b_3, b_4, b_5 are in G.P., b_4, b_5, b_6 are in A.P., b_5, b_6, b_7 are in G.P. and so on. Also given that $b_1 = 1$ and $b_5 + b_6 = 198$. Then

	Column I		Column II
(P)	$\sqrt{b_7}$ is equal to	(1)	5
(Q)	Sum of digits of b_8 is equal to	(2)	15
(R)	$\sqrt{b_9}$ is equal to	(3)	9
(S)	Sum of digits of b_{10} is equal to	(4)	17
		(5)	13
		(6)	14

- A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$ B) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$
 C) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$ D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

53.

	Column I		Column II
(P)	$2 \cot(\cot^{-1}(3) + \cot^{-1}(7) + \cot^{-1}(13) + \cot^{-1}(21))$ has the value equal to	(1)	3
(Q)	If $\tan\left(\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right) + \dots + \tan^{-1}\left(\frac{1}{381}\right)\right) = \frac{m}{n}$, where $m, n \in \mathbb{N}$, then the least value of $(m + n)$ is divisible by	(2)	4
(R)	Number of integral ordered pairs (x, y) satisfying the equation $\arctan \frac{1}{x} + \arctan \frac{1}{y} = \arctan \frac{1}{10}$, is	(3)	5
(S)	The smallest positive integral value of n for which $(n - 2)x^2 + 8x + n + 4 > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12)$ $\forall x \in \mathbb{R}$, is	(4)	8
		(5)	10

- A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$ B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$
 C) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$ D) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 4$

54.

Match List – I with List – II and select correct answer using the codes given below the list.

	Column I		Column II
(P)	$\int_0^{\pi/2} (e^{\sin x} - e^{\cos x}) dx$ equals	(1)	$10(1 - \cos 1)$
(Q)	$\int_0^{10} \sin \{x\} dx$ equals (where $\{ \cdot \}$ denotes fractional part function)	(2)	1
(R)	$\int_2^4 \frac{\sqrt{\ln(9-x)} dx}{\sqrt{\ln(9-x)} + \sqrt{\ln(x+3)}}$ equals	(3)	-1
(S)	$\frac{\int_0^1 x^5 (1-x^2)^7 dx}{\int_0^1 (x-1)^5 (2x-x^2)^7 dx}$ equals	(4)	0
		(5)	2
		(6)	-2

A) $P \rightarrow 4$; $Q \rightarrow 1$; $R \rightarrow 2$; $S \rightarrow 3$ B) $P \rightarrow 1$; $Q \rightarrow 4$; $R \rightarrow 3$; $S \rightarrow 2$ C) $P \rightarrow 3$; $Q \rightarrow 1$; $R \rightarrow 2$; $S \rightarrow 4$ D) $P \rightarrow 4$; $Q \rightarrow 1$; $R \rightarrow 3$; $S \rightarrow 2$